

Ben Bubnick

HEALTHCARE DATA STRATEGY · DATA GOVERNANCE · ANALYTICS DELIVERY

 bubnicbf |  benbubnick

Summary

Healthcare Data Consultant with 10+ years of experience leading the technical delivery and governance of analytics, data platforms, and ML solutions across clinical, operational, and revenue-cycle domains. Combines hands-on expertise in Python, SQL, Snowflake, SQL Server, AWS, Azure, and production data pipelines with leadership in data quality, source-to-target reconciliation, documentation, and shared standards. Skilled at turning ambiguous healthcare challenges into trusted analysis, reusable data models, real-time services, and stakeholder-ready reporting supported by testing, CI/CD, and observability. Known for aligning engineering, data science, operations, and business stakeholders while establishing transparent, scalable, and well-governed solutions for regulated environments.

Experience

HHaEXchange

Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase/GitHub Actions, FHIR/Aidbox

HEALTHCARE DATA INTEGRATION CONSULTANT — CONTRACT VIA VEERSA TECHNOLOGIES

Oct. 2025 - Jun. 2026

Led healthcare source analysis, data-quality investigation and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use.

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- Led source analysis and Snowflake/dbt delivery that converted operational healthcare data into standardized, analytics-ready models.
- Built reusable SQL/Jinja logic for lifecycle fields, null handling, sentinel dates and source-system variations across organization, office, caregiver and patient data.
- Developed QA suites for duplicates, nulls, referential integrity and reconciliation; analyzed failing source and target records to isolate discrepancies, quantify affected records and identify root causes.
- Assessed source structures and implementation constraints, working with client and internal stakeholders to clarify ambiguous business rules, define scope and establish clear, consistent and repeatable onboarding standards.

Adonis

AWS: S3, Snowflake, SQLMesh, SQL, Python

HEALTHCARE REVENUE CYCLE DATA CONSULTANT — CONTRACT VIA HEALTH DATA MOVERS

Mar. 2025 - Jul. 2025

Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding and reporting.

- Developed Snowflake/SQLMesh models for payer and revenue-cycle data, translating client requirements and source schemas into analysis-ready datasets.
- Built reusable transformations and validation checkpoints for row counts, nulls, referential integrity, and reconciliation.
- Added schema-drift detection, audit logs, CI checks, and deterministic recovery procedures to reduce regression risk and improve trust in delivered data.
- Produced mapping specifications and integration playbooks that made client onboarding logic transparent, repeatable and easier to review.

GrowData Analytics

Ruby, RubyLLM, Tesseract OCR, GPT 4o mini, Cladue Opus/Sonnet, Rubocop, RSpec

INDEPENDENT DATA AUTOMATION & ANALYTICS CONSULTANT

Sep. 2024 - Oct. 2025

Built a document-to-data product that converted semi-structured invoices into validated, auditable transaction records.

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- Built an OCR-to-analytics workflow that converted handwritten and printed invoices into structured transactions using cross-field validation, reconciliation and confidence-based exception handling.
- Produced audit logs, error taxonomies and usage reporting to investigate extraction failures, manage retries and improve processing reliability.

Arkos Health

AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter

SENIOR DATA ANALYST

Jun. 2023 - Jan. 2025

Delivered healthcare analytics, recurring reporting and data-quality investigations while improving the governed data foundation used by the BI team.

- Reconciled membership and demographic reporting discrepancies by isolating differences in extraction criteria, eligibility rules, distinct-member logic and duplicate handling; translated findings into stakeholder-ready BI reports.
- Established a governed repository for scheduled reports, ad hoc queries, tests and notebooks; introduced shared review standards, documented analytical methods and mentored analysts on collaborative delivery.

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Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps

SENIOR DATA SCIENTIST

Jul. 2022 - Mar. 2023

Led data validation, compliance and migration assurance for a regulated healthcare platform modernization.

- Built automated ETL regression and data-comparison tests that detected schema drift, transformation defects and fidelity issues during migration from legacy HDFS systems to Azure.
- Led HIPAA-compliant deidentification and validation across more than 200 million patient records, supporting HITRUST readiness and reducing audit risk.

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IBM

Hadoop/HDFS/CDH: Spark, Hive & Impala, Kafka, Python, JRuby, SQL, Pig, Docker + Gradle

SENIOR DATA SCIENTIST

Nov. 2021 - Jul. 2022

Delivered client analytics and large-scale migration validation for healthcare administrative data.

- Led delivery of tailored analytics and reporting for client administrative data, translating business requirements into validated analytical outputs.
- Developed migration-validation analyses covering more than 150 billion patient records, identifying data-fidelity issues before production transition.

SENIOR DATA ARCHITECT

Dec. 2020 - Nov. 2021

Led clinical-data integration and unstructured-text analysis at enterprise scale.

- Led integration of more than 100 billion structured and semi-structured clinical records for downstream analytics and application workflows.
- Applied NLP and topic modeling to nearly six million provider notes, extracting patterns from unstructured clinical text and documenting reusable analytical workflows.

DATA ARCHITECT

Nov. 2017 - Dec. 2020

Owned client-facing integration, discrepancy resolution and data-quality improvement across multi-organization healthcare networks.

- Developed standardized data-validation and review processes that reduced client-validation rework by 25% and mapping rework by 45%.
- Partnered with Client Delivery and engineering teams to investigate data and application discrepancies across a 73-organization healthcare network, tracking findings in Jira, clarifying source logic and coordinating fixes through production.

SOFTWARE DEVELOPER, DATA PLATFORM

Sep. 2016 - Nov. 2017

Built internal data-curation and knowledge-transfer tools that reduced costs and onboarding effort.

- Built a data-curation tool that replaced a \$30,000 annual contract and rewrote project documentation, reducing knowledge-transfer time by 80%.

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DATA SCIENTIST

Nov. 2015 - Sep. 2016

Coordinated client integration delivery, contractor work and process automation.

- Coordinated delivery across 36 client data-integration projects and managed 12 remote contractors, establishing expectations for reviews, quality and issue resolution.
- Built automation that eliminated 28% of unnecessary development work and created more than 125 knowledge-base articles—approximately 30% of the team's online content—to standardize delivery practices.

Explorys

Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins

DATA SCIENCE CONSULTANT — CONTRACT VIA TEKSYSTEMS

Jun. 2015 - Nov. 2015

Supported clinical-data integration and population analytics across more than 10 client data warehouses.

- Investigated production data issues and developed scheduling controls for reliable client ingestion across multiple environments.
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AmTrust Financial Services

VB.NET, ASP.NET, C#, CSS, Javascript, HTML5, SQL, PL/SQL

SOFTWARE ENGINEER

Mar. 2014 - Jun. 2015

Developed MVC components and web forms for insurance policy software.

- Led development of insurance policy software and a new product offering.
- Analyzed 65+ rating forms to identify and resolve performance bottlenecks.
- Introduced unit testing for multi-tier MVC components.

PPG Industries

VB.NET, C#, SQL, C++, FORTRAN

COLOR SCIENTIST

Apr. 2012 - Jun. 2013

Developed numerical methods and software for automotive color matching and measurement.

- Improved color-matching classification accuracy by 40% through advanced numerical methods.
- Built cross-instrument correlation software that improved measurement consistency and QA reliability.
- Developed formulation and pigment-analysis methods for an automated automotive color-matching facility.

Multiple Institutions

ADJUNCT PROFESSOR

Aug. 2010 - May. 2019

Taught physics & astronomy lecture series and lab courses.

- Taught physics and astronomy courses using interactive, peer-led, and just-in-time instructional methods.
- Produced more than 700 hours of recorded lectures and lab material for on-demand learning.
- Exceeded evaluation benchmarks by 16–17% year over year.

Professional Recognition

HDM, SPECIAL RECOGNITION (2025), Recognized for exceptional performance and commitment in client delivery.

Arkos, ARKOS HEALTH BI STAR (2023), Recognition for Achieving Expeditioner Rank, Salesforce Trailhead.

Merative, MANAGER RECOGNITION FOR TRANSITION LEADERSHIP (2022), Recognized for systems migration efforts.

IBM, GOLD CHAMPION (2018 - 2022), Recognized for 1,477 classroom hours delivered and 43 certificates earned.

IBM, LAB SERVICES AWARD (2018), Recognized for coordinating internal teams to deliver solutions on time.

IBM, EMINENCE AND EXCELLENCE AWARD (2017), Recognized for developing a new data curation tool.

PPG, COLOR LAB AUTOMATION AWARD (2012), Recognized for expanding data collection and analysis reporting.

Volunteer

HSEA

ETL BRIGADE

Feb. 2025 - May. 2025

Built a reusable PostgreSQL data model to help community health organizations prototype analytics tools for claims, visits, and eligibility.

JULY 29, 2026

CSforAll

INSTRUCTOR

Oct. 2018 - May. 2019

Our goal was to provide computer science instruction in all of the District's high schools, with a special emphasis on students who do not have access to a computer or a stable Internet connection at home.

HIT in the CLE

COACH

Mar. 2017 - Mar. 2019

We were using a data science competition to educate students understand the potential of a career in health information technology and to assist IT companies in Northeast Ohio in developing a talent pipeline.

Code for America

ETL GUILD

Jan. 2015 - Jun. 2015

The initiative was brought about to apply data science tools to improve and expand public record access.

Talks

AI Hackathon 2020: Surgery Scheduling Optimization

Dec. 2020

"An algorithm uses past utilization and current waitlist info to predict surgery duration.

IBM

REACH Initiative: IBM Watson Care Insights

2019

Analyze patient records and real-time data via HL7 feeds, integrate insights into physician's EHR workflow.

IBM

Content & Data Analytics Learning Exchange

2018

Introduction of implementation of machine learning algorithms using real-world health data analytics.

IBM

2018 Innovation Research Interchange Fall Networks Conference

Sep. 2018

Using NLP to extract insight from unstructured data and transform it from a data lake.

IRI

Cross Team Training Series

2016

A 7-part series on using Hadoop for data ingestion that shows advantages of data lakes.

EXPLORYS

Publications

Reproducible Postgres load of Tuva seed datasets

Aug. 2025

Reproducible Postgres load of Tuva seed datasets with robust validation.

HSEA

Surgical Scheduling Optimization and Procedure Duration Prediction ^a

Dec. 2020

Improvement on traditional fixed ratio methods for total procedure time estimation.

IBM

Process and Requirements for CCD Data Integration

Oct. 2017

Processes and requirements for overcoming the many difficulties with CCD data curation.

IBM

AI Paint Formula Prediction from Spectroscopic Measurements ^b

May. 2015

Naive bayesian color and effects matching algorithm of complex automotive paint formula.

SHINYAPPS.IO

Characterization of Surface Coatings using RT Models for Color Matching

Feb. 2013

Approximation of Chandrasekhar RT model for atmospheric scattering applied to color matching.

USPTO

Massive Stellar Clusters in the Disk of the Milky Way Galaxy ^c

Dec. 2010

Spectroscopic and Photometric analysis of two open clusters in the galactic plane.

OHIOLINK ETD

VizieR Online Data Catalog: JHK photometry in Cluster [BDS2003] 107

May. 2008

Spectra were obtained 2005 June 1415 at the 3.0 m NASA Infrared Telescope Facility (IRTF) on Mauna Kea, Hawaii, using SpeX.

VIZIER

Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster [BDS2003] 107

Feb. 2008

New near-infrared classification spectra for nine candidate members of the galactic cluster [BDS2003] 107 (Cluster 107).

PASP

A Novel Method for Low-Toxic Photo-Etch Printing ^d

May. 2003

Application of aquatint etching techniques to photo-etch procedures.

MAPC

Education

2022 **IBM Data Science Professional Certificate**^a
2015 **Certificate of Specialization** Data Science^b
2010 **Master of Science** Astrophysics^c
2004 **Bachelor of Fine Arts** Printmaking^d

IBM
Johns Hopkins University
University of Cincinnati
Ohio University